

Newman-Penrose Formalism Calculations

Stephani et al., Equation (12.19)

Metric

$$ds^2 = \left(-\frac{k^2 + z^2}{k^2} \right) dt \otimes dt + \left(\frac{k^2 - y^2}{k^2} \right) dx \otimes dx + \left(\frac{k^2}{k^2 - y^2} \right) dy \otimes dy + \left(\frac{k^2}{k^2 + z^2} \right) dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{\frac{1}{2}} \sqrt{k^2 + z^2}}{k} \right) dt + \left(\frac{\sqrt{\frac{1}{2}} \sqrt{k^2 - y^2}}{k} \right) dx$$

$$n = \left(\frac{\sqrt{\frac{1}{2}} \sqrt{k^2 + z^2}}{k} \right) dt + \left(-\frac{\sqrt{\frac{1}{2}} \sqrt{k^2 - y^2}}{k} \right) dx$$

$$m = \left(\frac{\sqrt{2}k}{2\sqrt{k^2 - y^2}} \right) dy + \left(\frac{i\sqrt{2}k}{2\sqrt{k^2 + z^2}} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2}k}{2\sqrt{k^2 - y^2}} \right) dy + \left(-\frac{i\sqrt{2}k}{2\sqrt{k^2 + z^2}} \right) dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}k}{2\sqrt{k^2+z^2}} \right) dt + \left(\frac{\sqrt{2}k}{2\sqrt{k+y}\sqrt{k-y}} \right) dx$$

$$n = \left(-\frac{\sqrt{2}k}{2\sqrt{k^2+z^2}} \right) dt + \left(-\frac{\sqrt{2}k}{2\sqrt{k+y}\sqrt{k-y}} \right) dx$$

$$m = \left(\frac{\sqrt{2}\sqrt{k+y}\sqrt{k-y}}{2k} \right) dy + \left(\frac{i\sqrt{2}\sqrt{k^2+z^2}}{2k} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2}\sqrt{k+y}\sqrt{k-y}}{2k} \right) dy + \left(-\frac{i\sqrt{2}\sqrt{k^2+z^2}}{2k} \right) dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2}k}{2\sqrt{k^2+z^2}} \partial_t + \frac{\sqrt{2}k}{2\sqrt{k+y}\sqrt{k-y}} \partial_x$$

$$\Delta = -\frac{\sqrt{2}k}{2\sqrt{k^2+z^2}} \partial_t - \frac{\sqrt{2}k}{2\sqrt{k+y}\sqrt{k-y}} \partial_x$$

$$\delta = \frac{\sqrt{2}\sqrt{k+y}\sqrt{k-y}}{2k} \partial_y + \frac{i\sqrt{2}\sqrt{k^2+z^2}}{2k} \partial_z$$

$$\bar{\delta} = \frac{\sqrt{2}\sqrt{k+y}\sqrt{k-y}}{2k} \partial_y - \frac{i\sqrt{2}\sqrt{k^2+z^2}}{2k} \partial_z$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = -\frac{\sqrt{2}(i\sqrt{k+y}\sqrt{k-y}z + \sqrt{k^2+z^2}y)}{4\sqrt{k^2+z^2}\sqrt{k+y}\sqrt{k-y}k}$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{2}(-i\sqrt{k+y}\sqrt{k-y}z + \sqrt{k^2+z^2}y)}{4\sqrt{k^2+z^2}\sqrt{k+y}\sqrt{k-y}k}$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = \frac{\sqrt{2}(-i\sqrt{k+y}\sqrt{k-y}z + \sqrt{k^2+z^2}y)}{4\sqrt{k^2+z^2}\sqrt{k+y}\sqrt{k-y}k}$$

$$\pi = -\frac{\sqrt{2}(i\sqrt{k+y}\sqrt{k-y}z + \sqrt{k^2+z^2}y)}{4\sqrt{k^2+z^2}\sqrt{k+y}\sqrt{k-y}k}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = \frac{1}{2k^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = \frac{1}{2k^2}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = \frac{1}{2k^2}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{1}{2k^2}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "O"